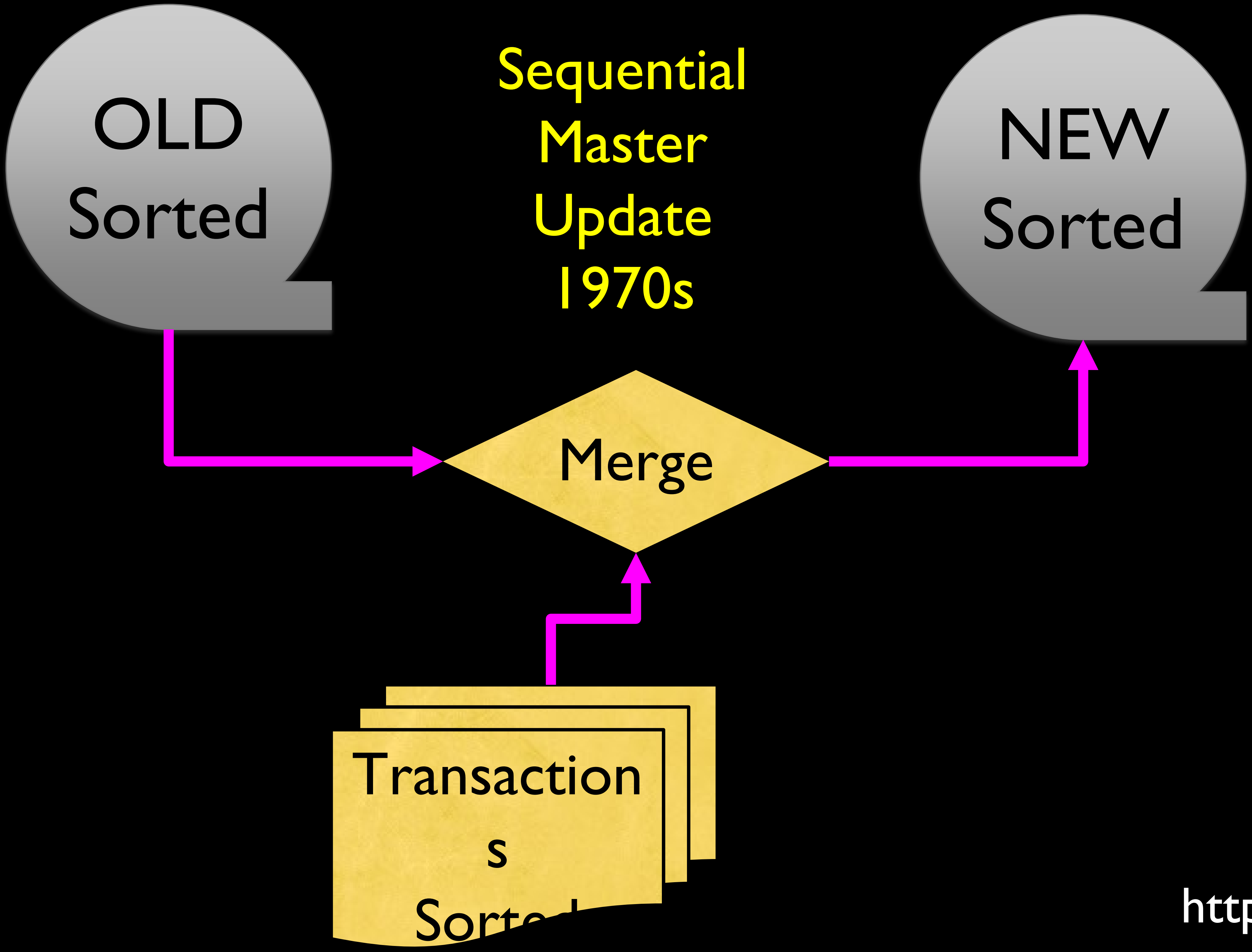
Relational Databases and MySQL

Charles Severance
www.wa4e.com

Sequential Master Update 1970s



https://en.wikipedia.org/wiki/IBM_729

Random Access

- When you can randomly access data...
- How can you lay out data to be most efficient?
- Sorting might not be the best idea

https://en.wikipedia.org/wiki/Hard_disk_drive_platter

Relational Databases

Relational databases model data by storing rows and columns in tables. The power of the relational database lies in its ability to efficiently retrieve data from those tables - in particular, where the query involves multiple tables and the relationships between those tables.

http://en.wikipedia.org/wiki/Relational_database

Structured Query Language

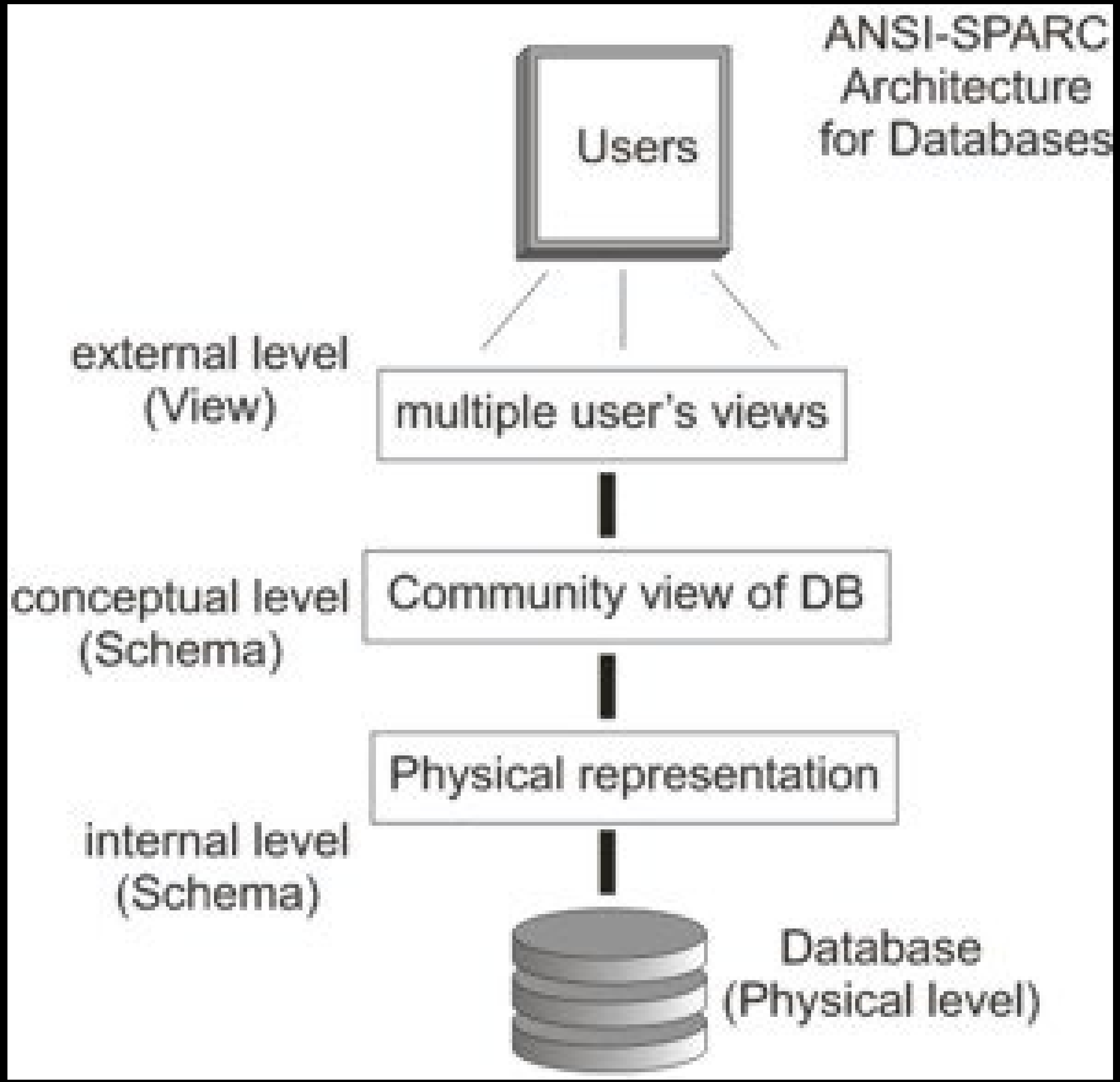
- Structured Query Language (SQL) came out of a government / industry partnership
- National Institute of Standards and Technology (NIST)

https://youtu.be/rLUm3vst87g

SQL

Structured Query Language is the language we use to issue commands to the database

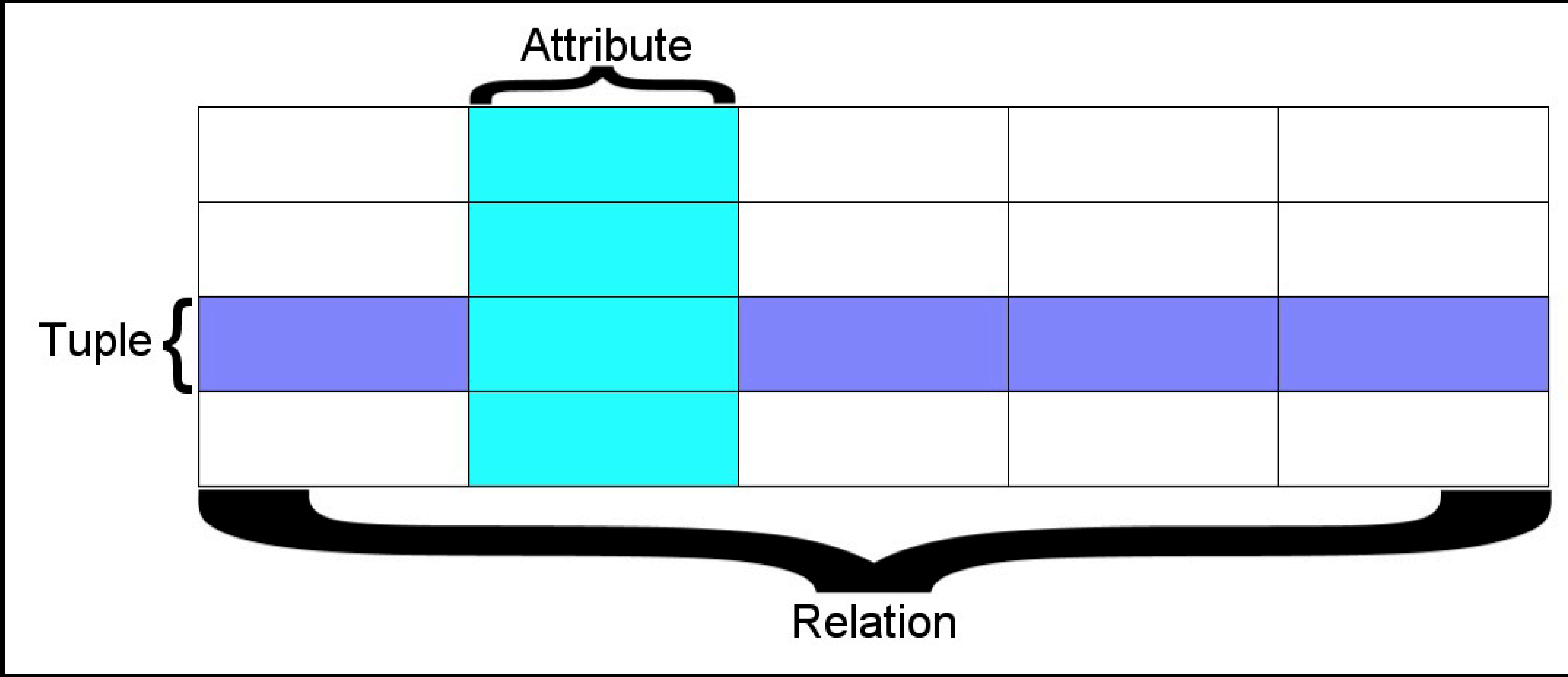
- Create/Insert data
- Read/Select some data
- Update data
- Delete data



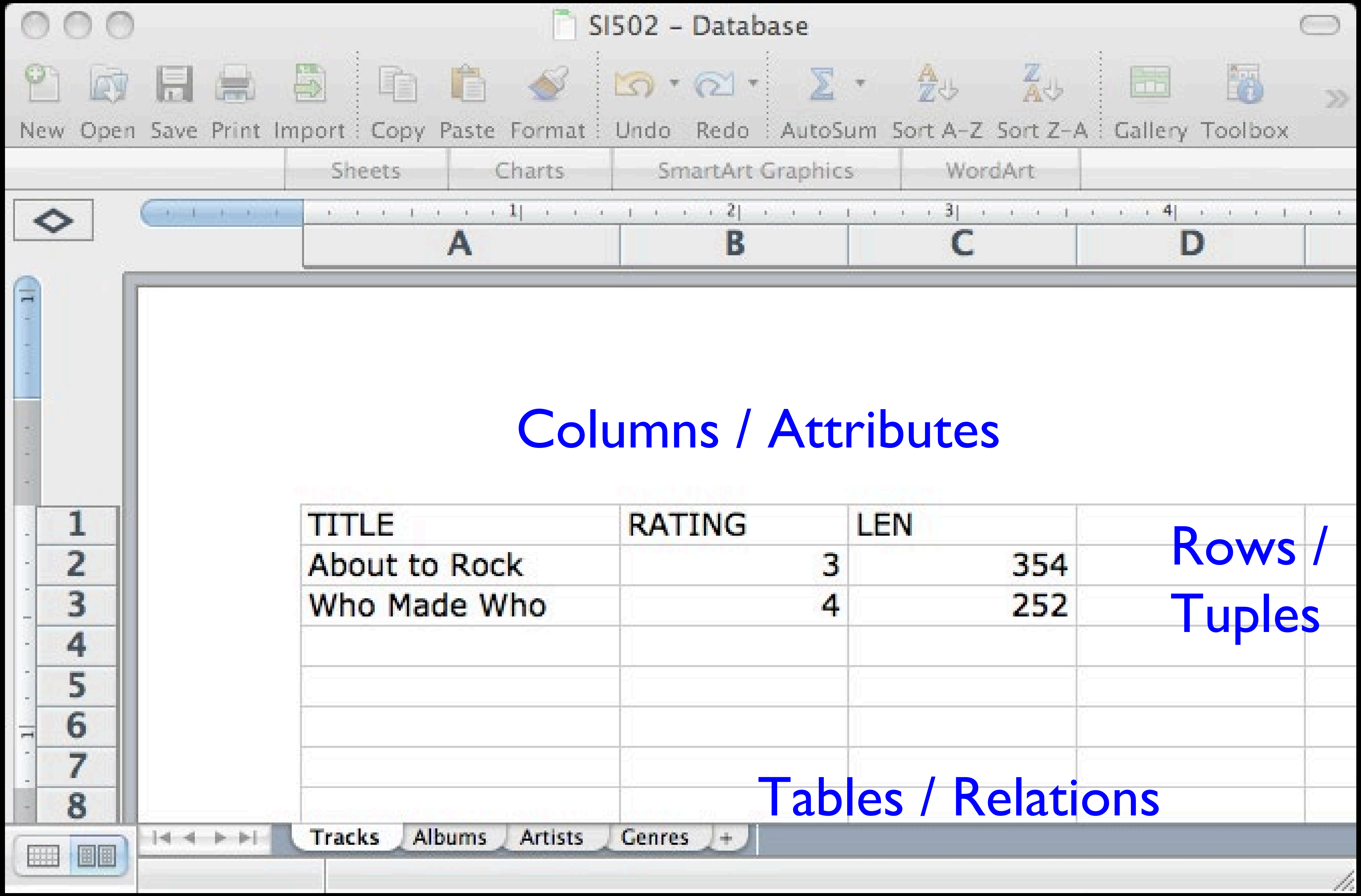
http://en.wikipedia.org/wiki/SQL
https://en.wikipedia.org/wiki/ANSI-SPARC_Architecture

Terminology

- Database - contains one or more tables
- Relation (or table) - contains tuples and attributes
- Tuple (or row) - a set of fields which generally represent an "object" like a person or a music track
- Attribute (also column or field) - one of possibly many elements of data corresponding to the object represented by the row



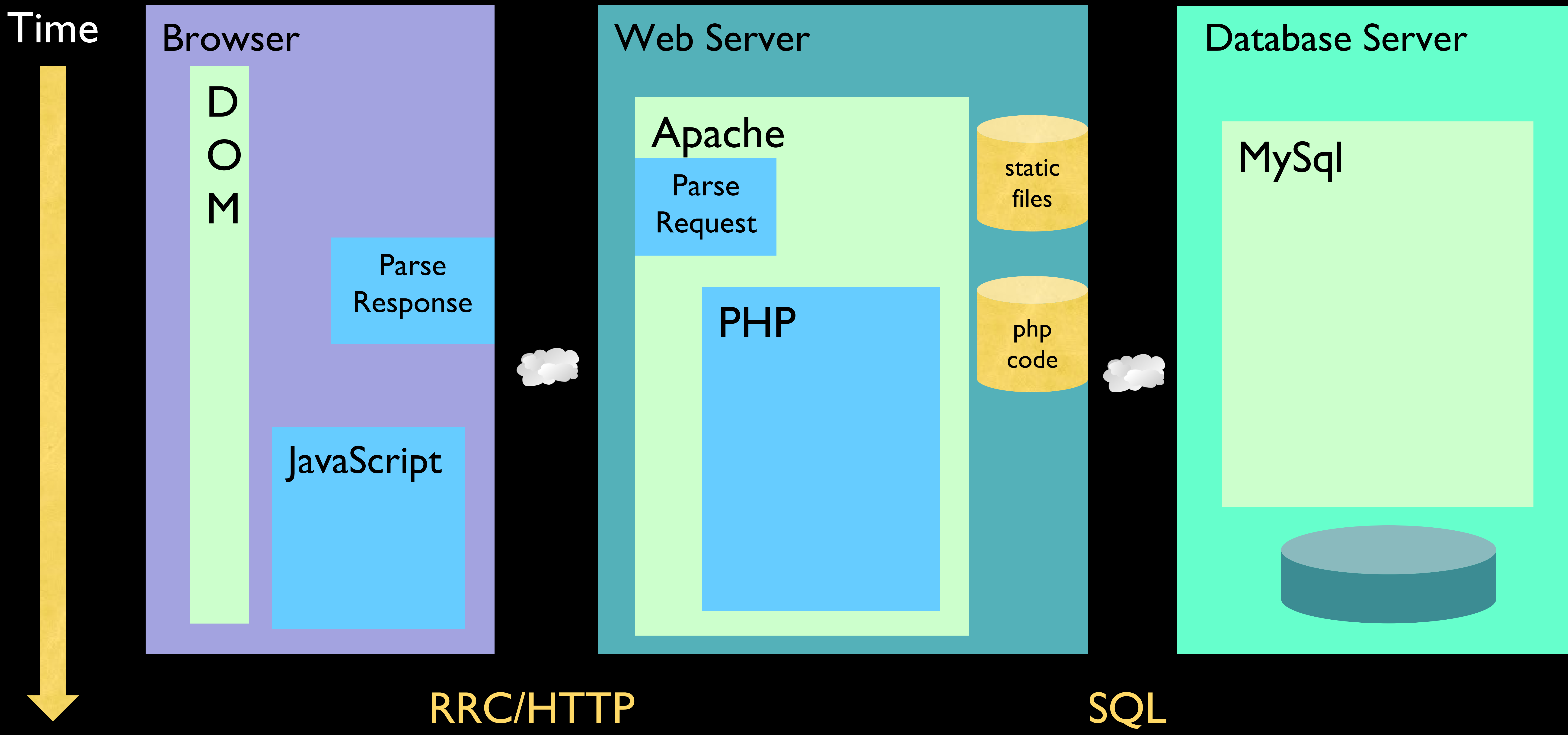
A relation is defined as a set of tuples that have the same attributes. A tuple usually represents an object and information about that object. Objects are typically physical objects or concepts. A relation is usually described as a table, which is organized into rows and columns. All the data referenced by an attribute are in the same domain and conform to the same constraints. (wikipedia)


SI502 – Database
New
Open
Save
Print
Import
Copy
Paste
Format
Undo
Redo
AutoSum
Sort A-Z
Sort Z-A
Gallery
Toolbox
Sheets
Charts
SmartArt Graphics
WordArt
A
B
C
D
Columns / Attributes
TITLE
RATING
LEN
About to Rock
3
354
Who Made Who
4
252
Rows / Tuples
Tables / Relations
Tracks
Albums
Artists
Genres

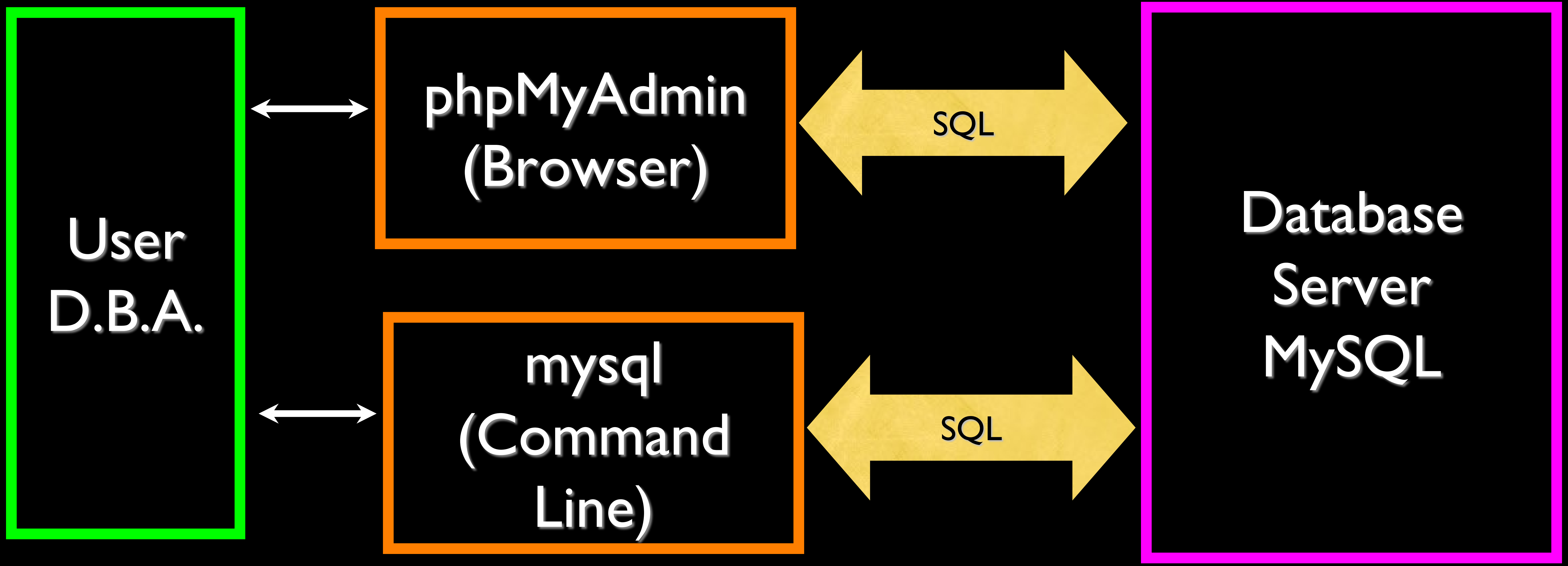
Common Database Systems

- Three major Database Management Systems in wide use
  - Oracle - Large, commercial, enterprise-scale, very tweakable
  - MySQL - Simpler but very fast and scalable - commercial open source
  - SqlServer - Very nice - from Microsoft (also Access)
- Many other smaller projects, free and open source
  - HSQL, SQLite, PostgreSQL ...

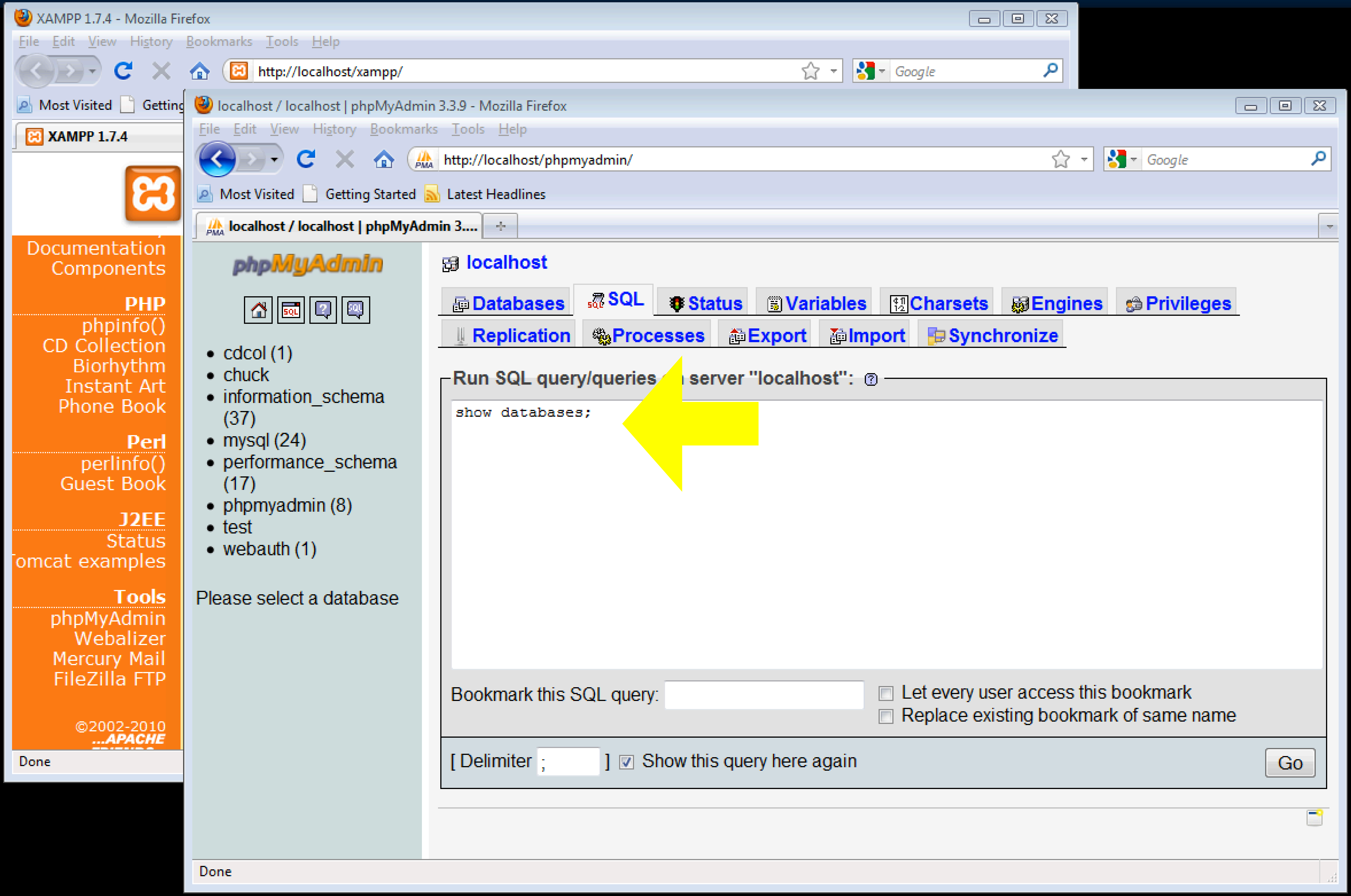
Basic SQL Operations

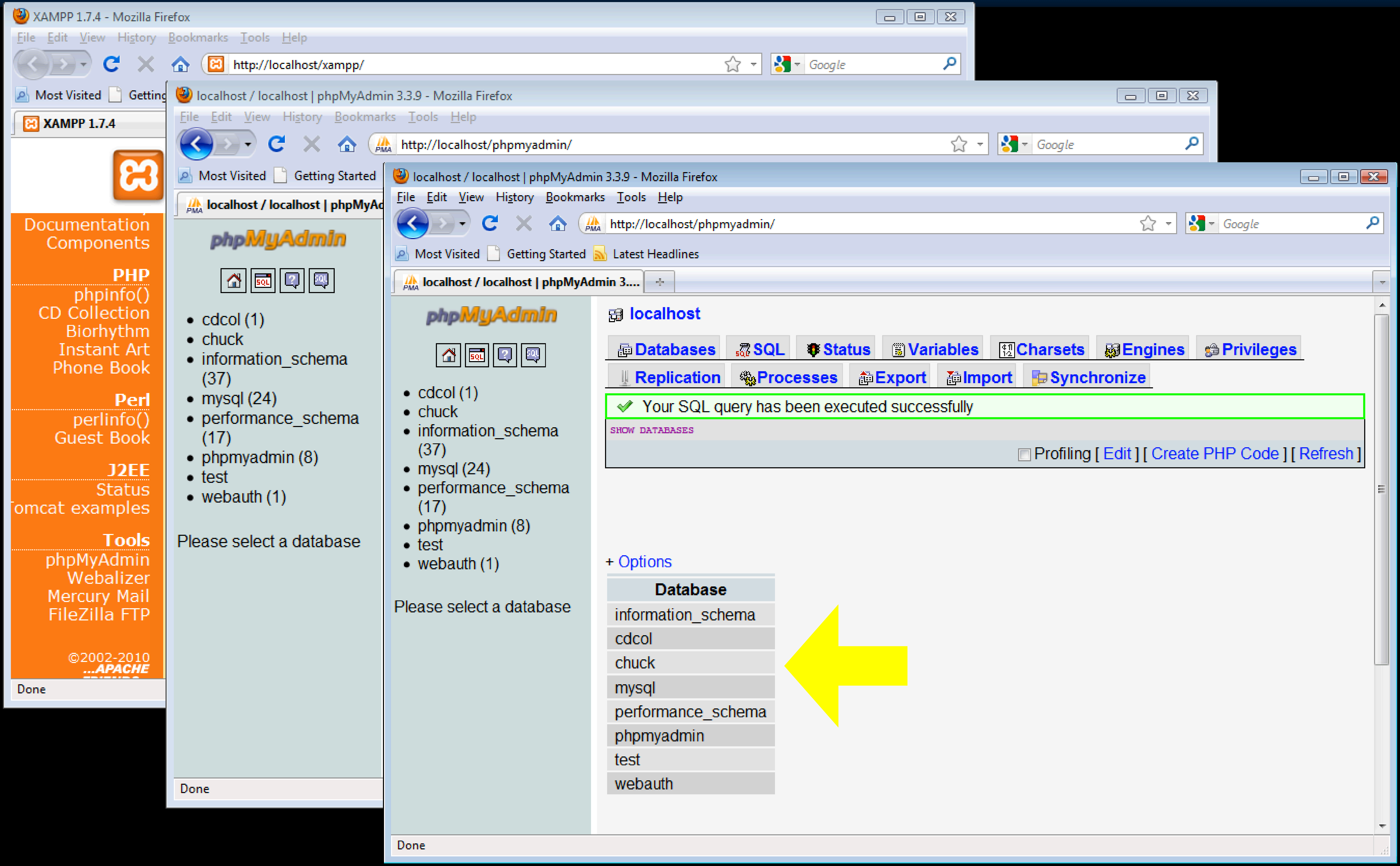

Time
Browser
DOM
Parse Response
JavaScript
Web Server
Apache
Parse Request
PHP
static files
php code
Database Server
MySql
RRC/HTTP
SQL

Using SQL


User
D.B.A.
phpMyAdmin
(Browser)
mysql
(Command
Line)
SQL
SQL
Database
Server
MySQL

Rational Databases and MySQL


XAMPP 1.7.4 - Mozilla Firefox
http://localhost/xampp/
Documentation
Components
PHP
phpinfo()
CD Collection
Biorhythm
Instant Art
Phone Book
Perl
perlinfo()
Guest Book
J2EE
Status
Tools
phpMyAdmin
Webalizer
Mercury Mail
FileZilla FTP
localhost / localhost | phpMyAdmin 3.3.9 - Mozilla Firefox
http://localhost/phpmyadmin/
Most Visited
Getting Started
Latest Headlines
phpMyAdmin
cdcol (1)
chuck
information_schema (37)
mysql (24)
performance_schema (17)
phpmyadmin (8)
test
webauth (1)
Please select a database
localhost
Databases
SQL
Status
Variables
Charsets
Engines
Privileges
Replication
Processes
Export
Import
Synchronize
Run SQL query/queries on server "localhost":
show databases;
Bookmark this SQL query:
Let every user access this bookmark
Replace existing bookmark of same name
[Delimiter ;]
Show this query here again
Go
Done


localhost / localhost | phpMyAdmin 3.3.9 - Mozilla Firefox
http://localhost/phpmyadmin/
localhost
Databases
SQL
Status
Variables
Charsets
Engines
Privileges
Replication
Processes
Export
Import
Synchronize
Your SQL query has been executed successfully
SHOW DATABASES
Profiling [Edit] [Create PHP Code] [Refresh]
+ Options
Database
information_schema
cdcol
chuck
mysql
performance_schema
phpmyadmin
test
webauth
Please select a database
XAMPP 1.7.4

Command Line

After Control Panel is running...

- Macintosh
  - /Applications/MAMP/Library/bin/mysql --host=localhost -uroot –p
  - Enter "root" when prompted for the password
- Windows
  - c:\xampp\mysql\bin\mysql.exe -u root –p
  - Press enter when prompted for password

Rational Databases and MySQL

Administrator: C:\Windows\system32\cmd.exe - c:\xampp\mysql\bin\mysql.exe -u root

```
C:\Users\Administrator>c:\xampp\mysql\bin\mysql.exe -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 3
Server version: 5.5.8 MySQL Community Server (GPL)

Copyright (c) 2000, 2010, Oracle and/or its affiliates. All rights reserved.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Administrator: C:\Windows\system32\cmd.exe - c:\xampp\mysql\bin\mysql.exe -u root

```
C:\Users\Administrator>c:\xampp\mysql\bin\mysql.exe -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 3
Server version: 5.5.8 MySQL Community Server (GPL)

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affiliates. Oth
owners.

Type 'he

mysql>
```

csev — mysql — 94×23

mysql

```
cseverancembp:~ csev$ /Applications/MAMP/Library/bin/mysql --host=localhost -uroot -proot
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 4
Server version: 5.5.25 Source distribution

Copyright (c) 2000, 2011, Oracle and/or its affiliates. All rights reserved.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Your First MySQL Command

show databases

Kind of like print('hello world’)

Default

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| cdcol              |
| chuck              |
| mysql              |
| sakai              |
| test               |
+--------------------+
6 rows in set (0.06 sec)

mysql> 
```

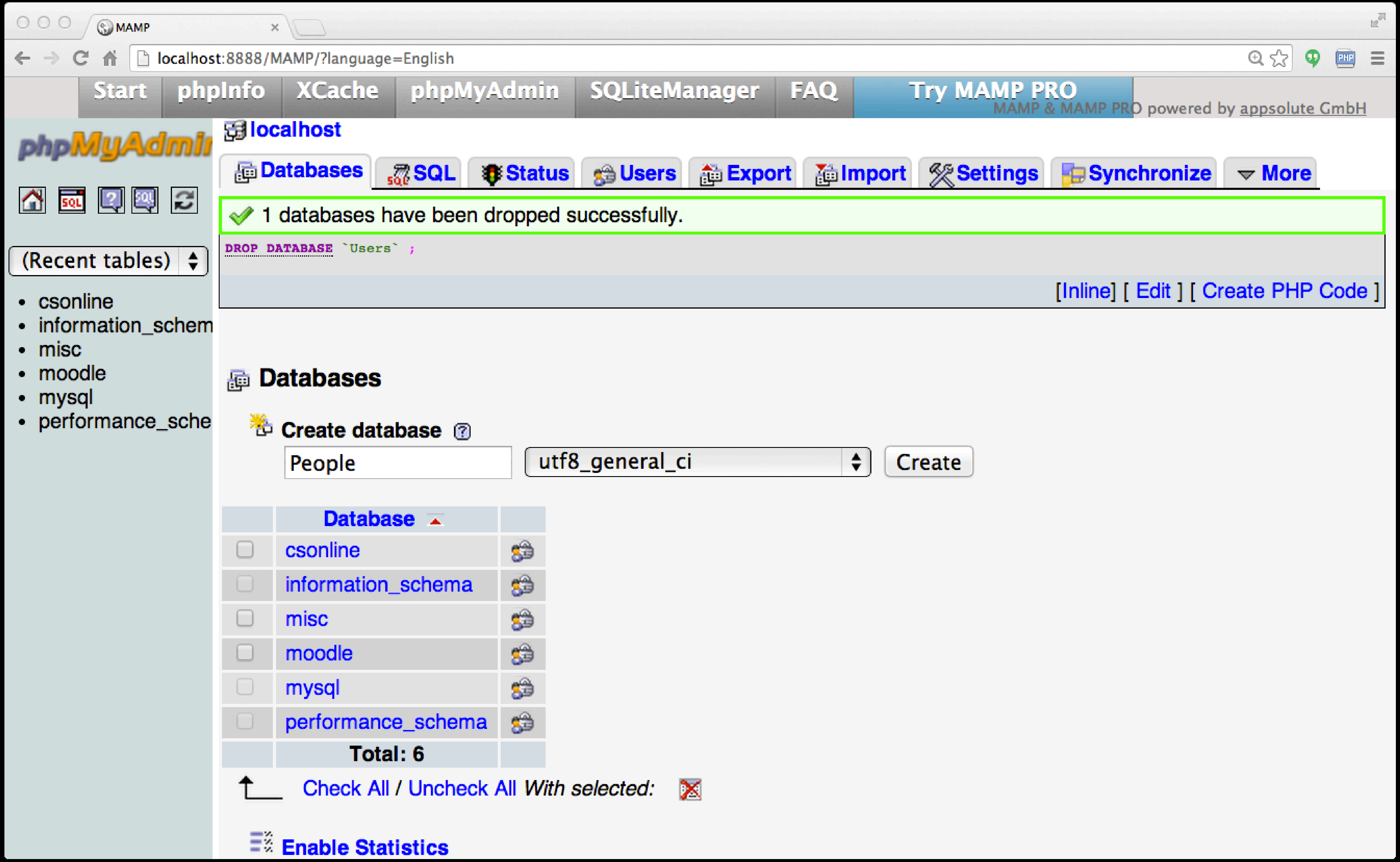
If this does not work, stop and figure out why.

Some of these are part of MySQL and store internal data - don't mess with them.

Creating a Database

Command Line:

```
CREATE DATABASE People;
USE People;
```

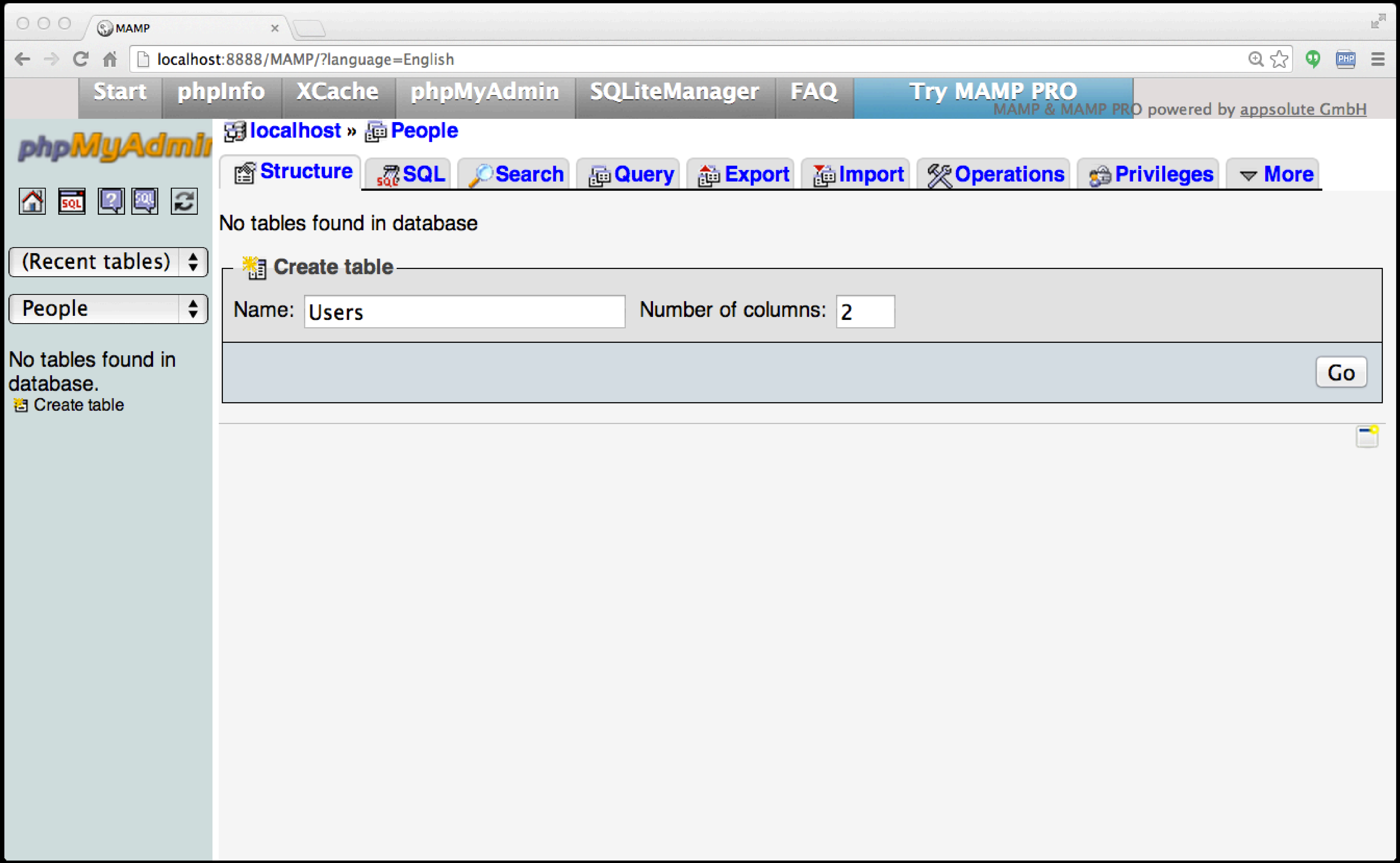

MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
localhost
Databases
SQL
Status
Users
Export
Import
Settings
Synchronize
More
1 databases have been dropped successfully.
DROP DATABASE `Users` ;
[Inline] [Edit] [Create PHP Code]
(Recent tables)
csonline
information_schem
misc
moodle
mysql
performance_sche
Databases
Create database
People
utf8_general_ci
Create
Database
csonline
information_schema
misc
moodle
mysql
performance_schema
Total: 6
Check All / Uncheck All With selected:
Enable Statistics

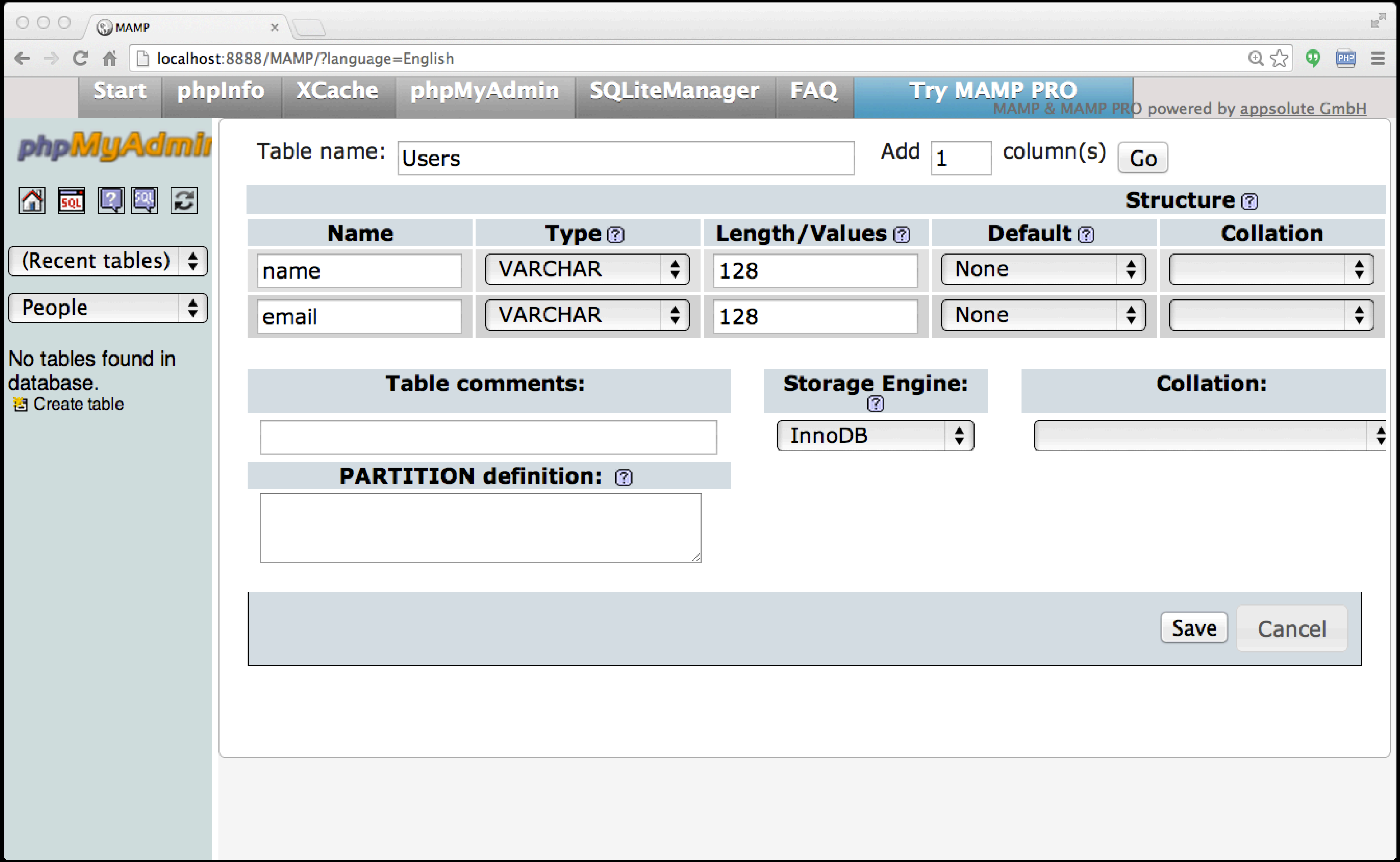
Start Simple - A Single Table

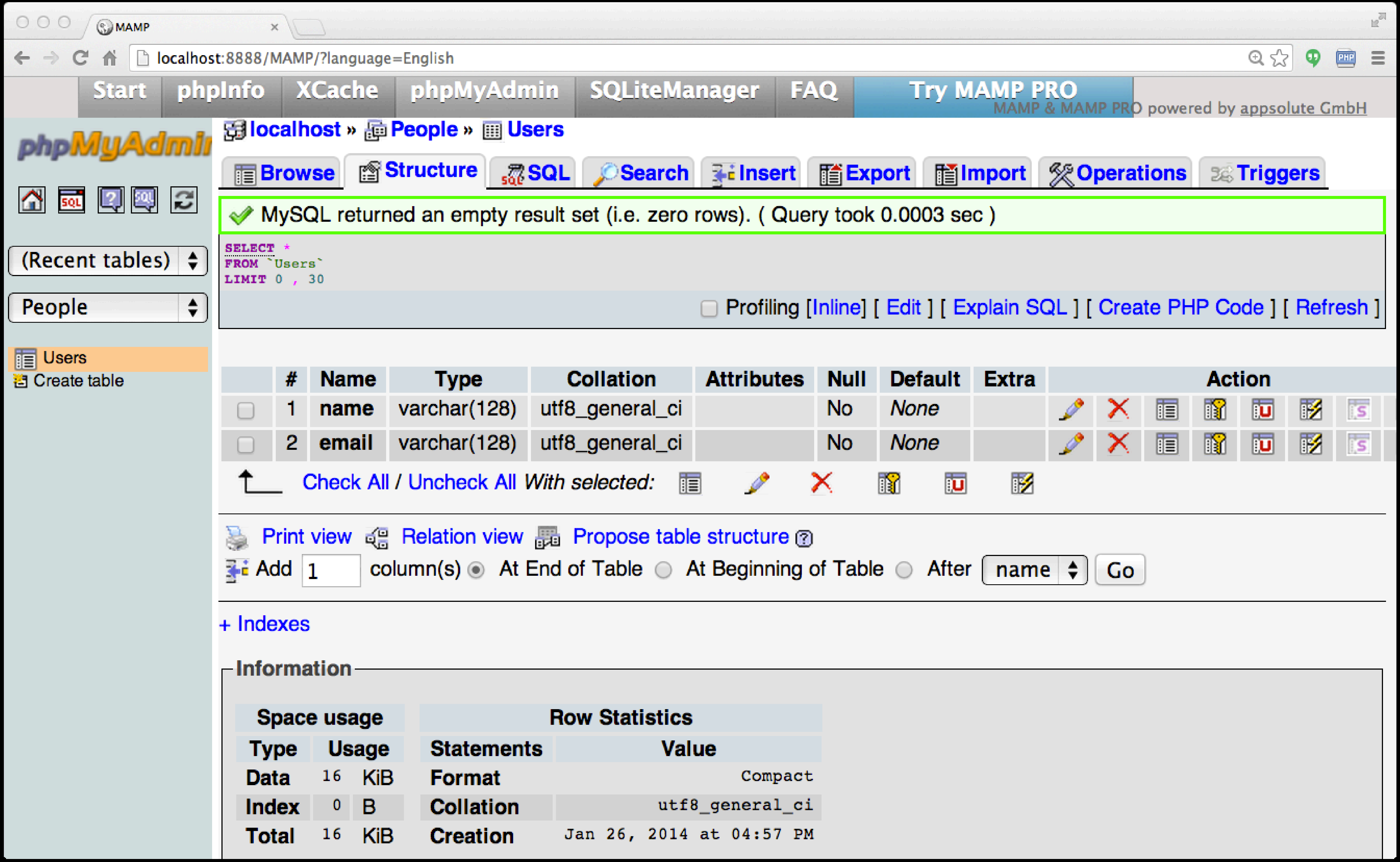
- Let’s make a table of Users in our People database
- Two columns - name and email

```
CREATE TABLE Users(
  name VARCHAR(128),
  email VARCHAR(128)
);

DESCRIBE Users;
```


MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
localhost » People
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
More
No tables found in database
Create table
Name: Users
Number of columns: 2
Go
(Recent tables)
People
No tables found in database.
Create table

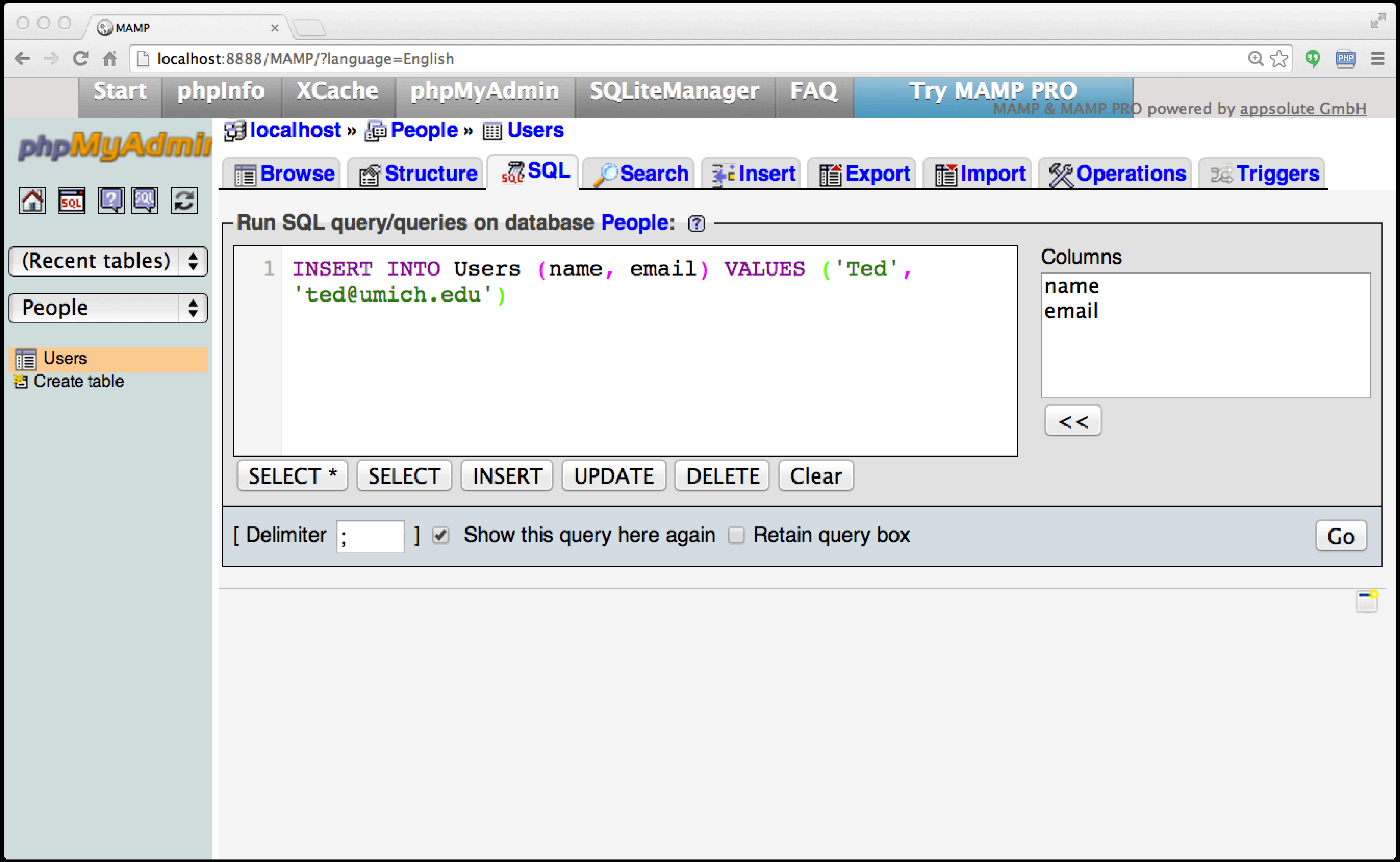

MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
(Recent tables)
People
No tables found in database.
Create table
Table name: Users
Add 1 column(s) Go
Structure
Name
Type
Length/Values
Default
Collation
name
VARCHAR
128
None
email
VARCHAR
128
None
Table comments:
Storage Engine:
InnoDB
Collation:
PARTITION definition:
Save
Cancel

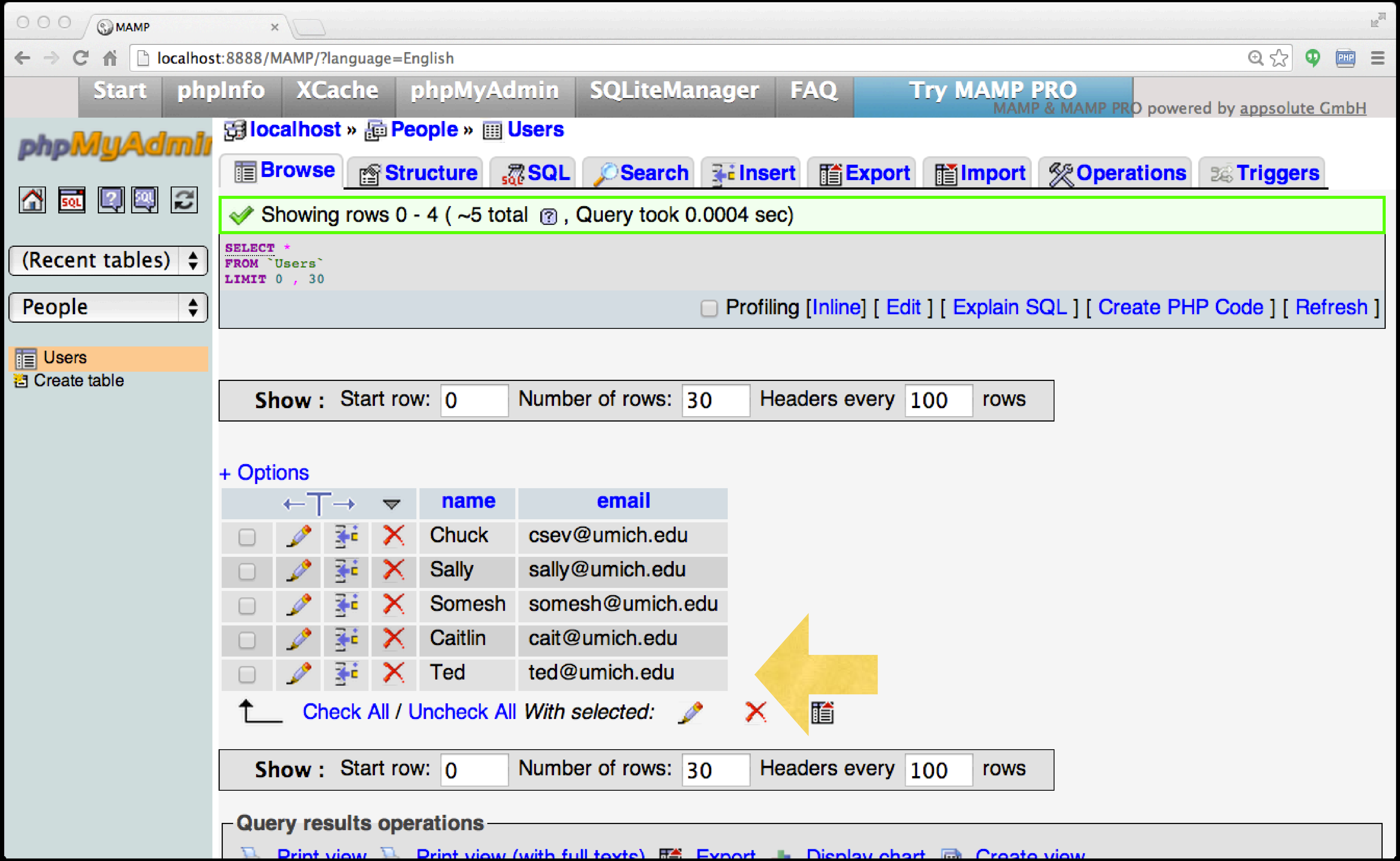

MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
(Recent tables)
People
Users
Create table
localhost » People » Users
Browse
Structure
SQL
Search
Insert
Export
Import
Operations
Triggers
MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 sec)
SELECT *
FROM `Users`
LIMIT 0 , 30
Profiling [Inline] [Edit] [Explain SQL] [Create PHP Code] [Refresh]
Name Type Collation Attributes Null Default Extra Action
1 name varchar(128) utf8_general_ci No None
2 email varchar(128) utf8_general_ci No None
Check All / Uncheck All With selected:
Print view
Relation view
Propose table structure
Add 1 column(s) At End of Table At Beginning of Table After name Go
+ Indexes
Information
Space usage
Type Usage
Data 16 KiB
Index 0 B
Total 16 KiB
Row Statistics
Statements Value
Format Compact
Collation utf8_general_ci
Creation Jan 26, 2014 at 04:57 PM

SQL: Insert

The INSERT statement inserts a row into a table

```
INSERT INTO Users (name, email) VALUES ('Chuck', 'csev@umich.edu') ;
INSERT INTO Users (name, email) VALUES ('Somesh', 'somesh@umich.edu') ;
INSERT INTO Users (name, email) VALUES ('Caitlin', 'cait@umich.edu') ;
INSERT INTO Users (name, email) VALUES ('Ted', 'ted@umich.edu') ;
INSERT INTO Users (name, email) VALUES ('Sally', 'sally@umich.edu') ;
```

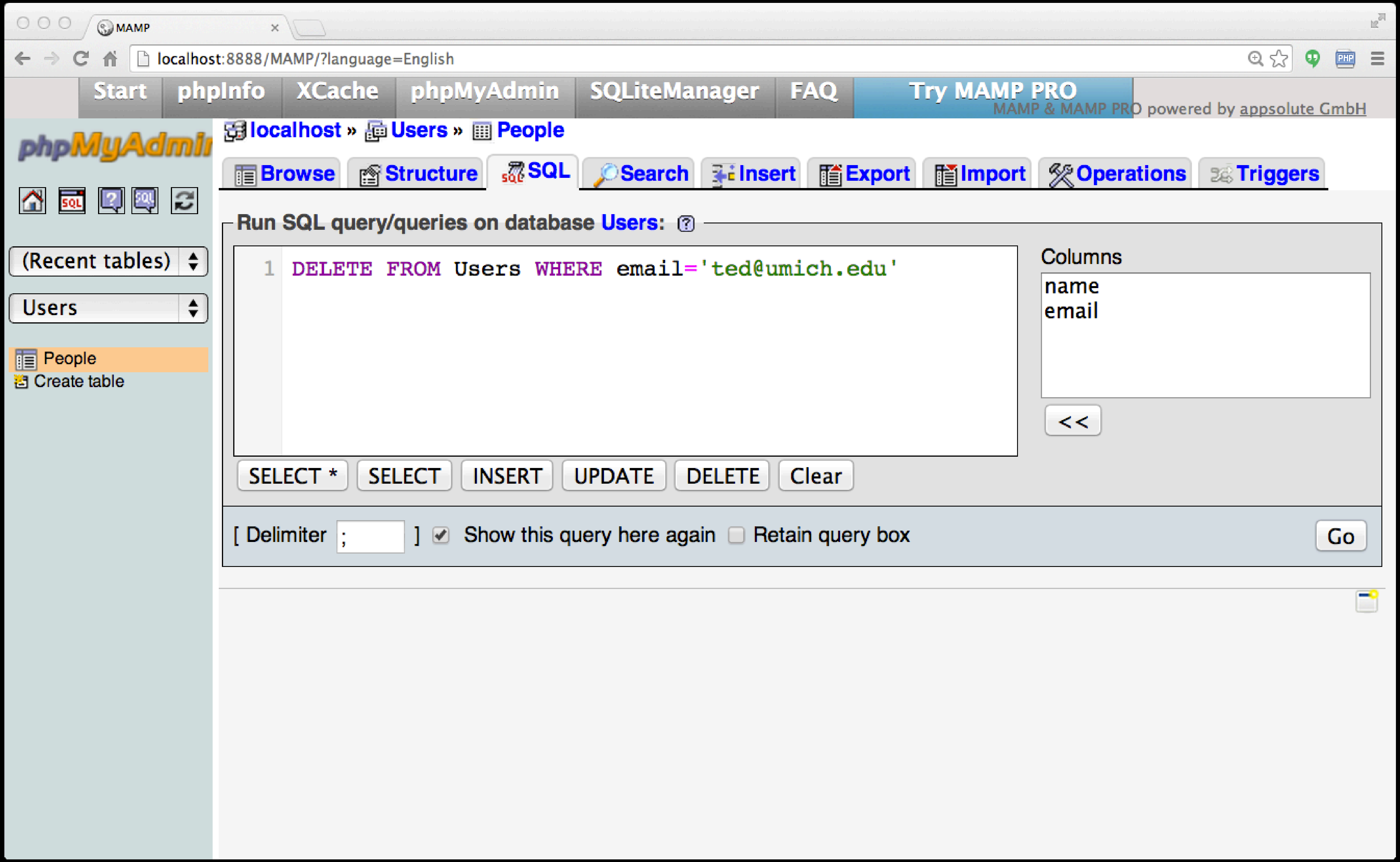

MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
localhost » People » Users
Browse
Structure
SQL
Search
Insert
Export
Import
Operations
Triggers
(Recent tables)
People
Users
Create table
Run SQL query/queries on database People:
INSERT INTO Users (name, email) VALUES ('Ted', 'ted@umich.edu')
Columns
name
email
<<
SELECT *
SELECT
INSERT
UPDATE
DELETE
Clear
[Delimiter ;]
Show this query here again
Retain query box
Go


MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
localhost » People » Users
Browse
Structure
SQL
Search
Insert
Export
Import
Operations
Triggers
Showing rows 0 - 4 (~5 total , Query took 0.0004 sec)
SELECT *
FROM `Users`
LIMIT 0 , 30
Profiling [Inline] [Edit] [Explain SQL] [Create PHP Code] [Refresh]
(Recent tables)
People
Users
Create table
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
+ Options
name
email
Chuck
csev@umich.edu
Sally
sally@umich.edu
Somesh
somesh@umich.edu
Caitlin
cait@umich.edu
Ted
ted@umich.edu
Check All / Uncheck All With selected:
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
Query results operations
Print view
Print view (with full texts)
Export
Display chart
Create view

SQL: Delete

Deletes a row in a table based on selection criteria

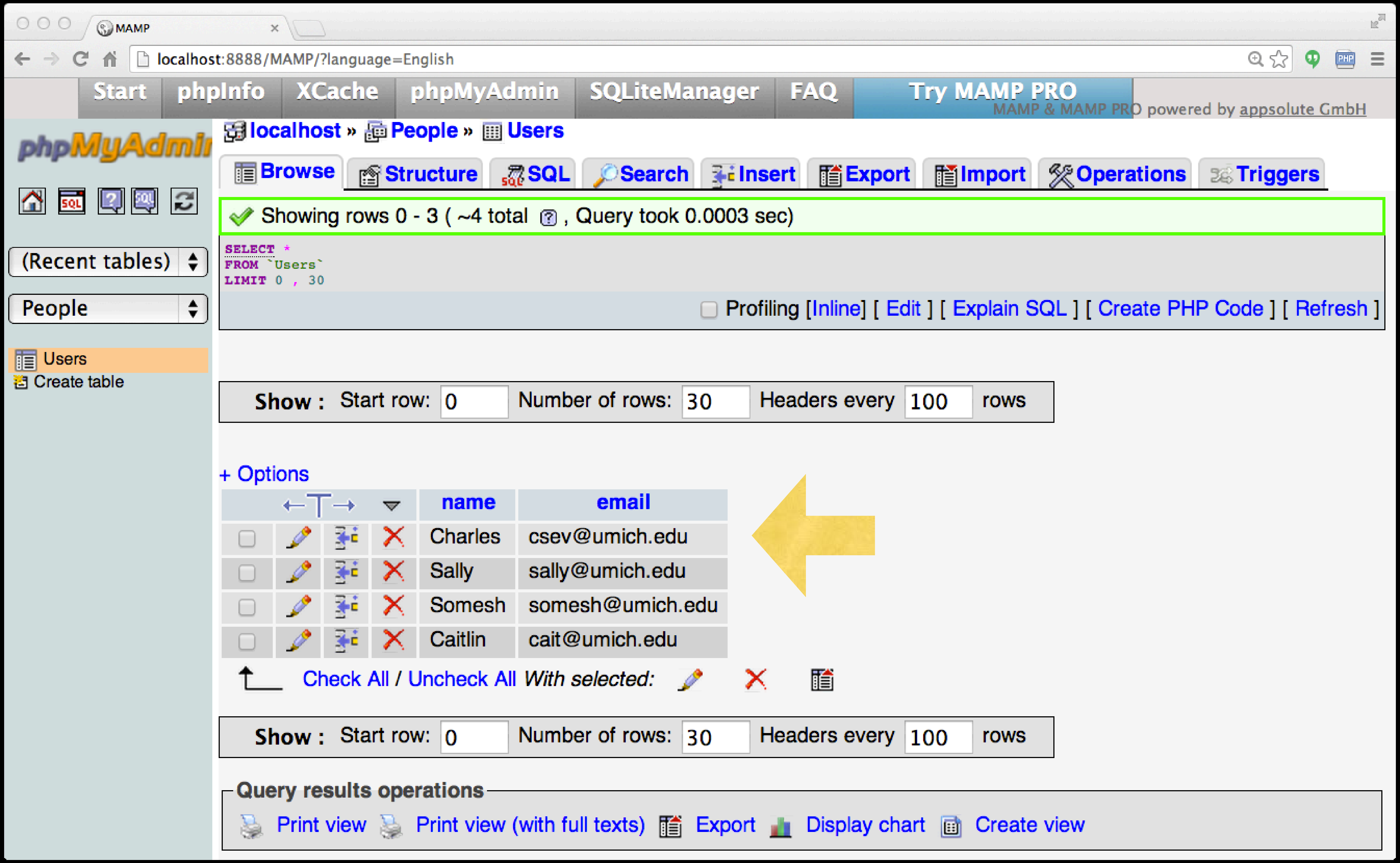
```
DELETE FROM Users WHERE email='ted@umich.edu'
```


MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
(Recent tables)
Users
People
Create table
localhost » Users » People
Browse
Structure
SQL
Search
Insert
Export
Import
Operations
Triggers
Run SQL query/queries on database Users:
1 DELETE FROM Users WHERE email='ted@umich.edu'
Columns
name
email
<<
SELECT *
SELECT
INSERT
UPDATE
DELETE
Clear
[Delimiter ;]
Show this query here again
Retain query box
Go

SQL: Update

Allows the updating of a field with a WHERE clause

```
UPDATE Users SET name='Charles' WHERE email='csev@umich.edu'
```

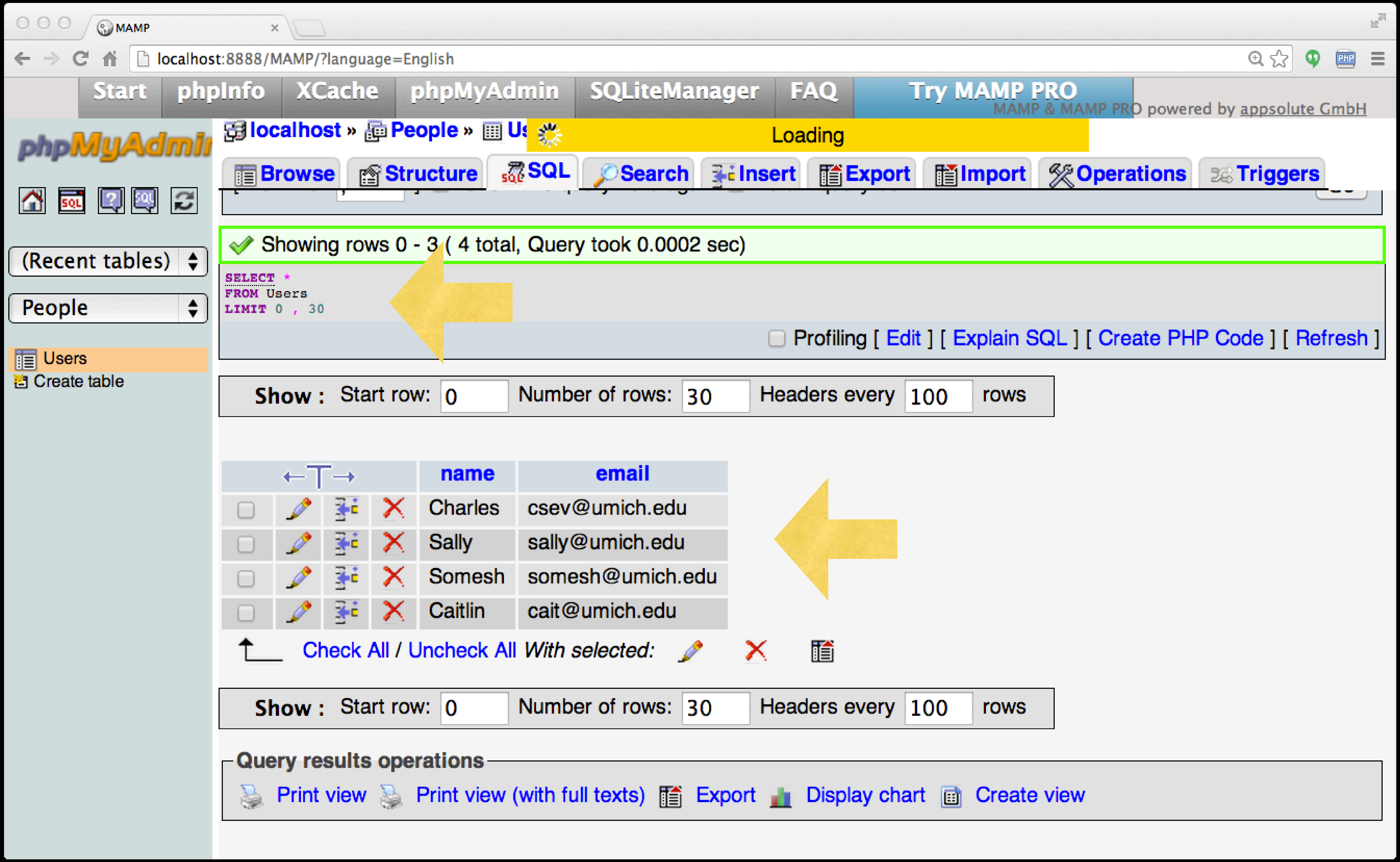

MAMP
localhost:8888/MAMP/?language=English
Start phpInfo XCache phpMyAdmin SQLiteManager FAQ Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
localhost » People » Users
Browse Structure SQL Search Insert Export Import Operations Triggers
Showing rows 0 - 3 (~4 total , Query took 0.0003 sec)
SELECT * FROM `Users` LIMIT 0 , 30
Profiling [Inline] [Edit] [Explain SQL] [Create PHP Code] [Refresh]
(Recent tables)
People
Users
Create table
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
+ Options
name email
Charles csev@umich.edu
Sally sally@umich.edu
Somesh somesh@umich.edu
Caitlin cait@umich.edu
Check All / Uncheck All With selected:
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
Query results operations
Print view Print view (with full texts) Export Display chart Create view

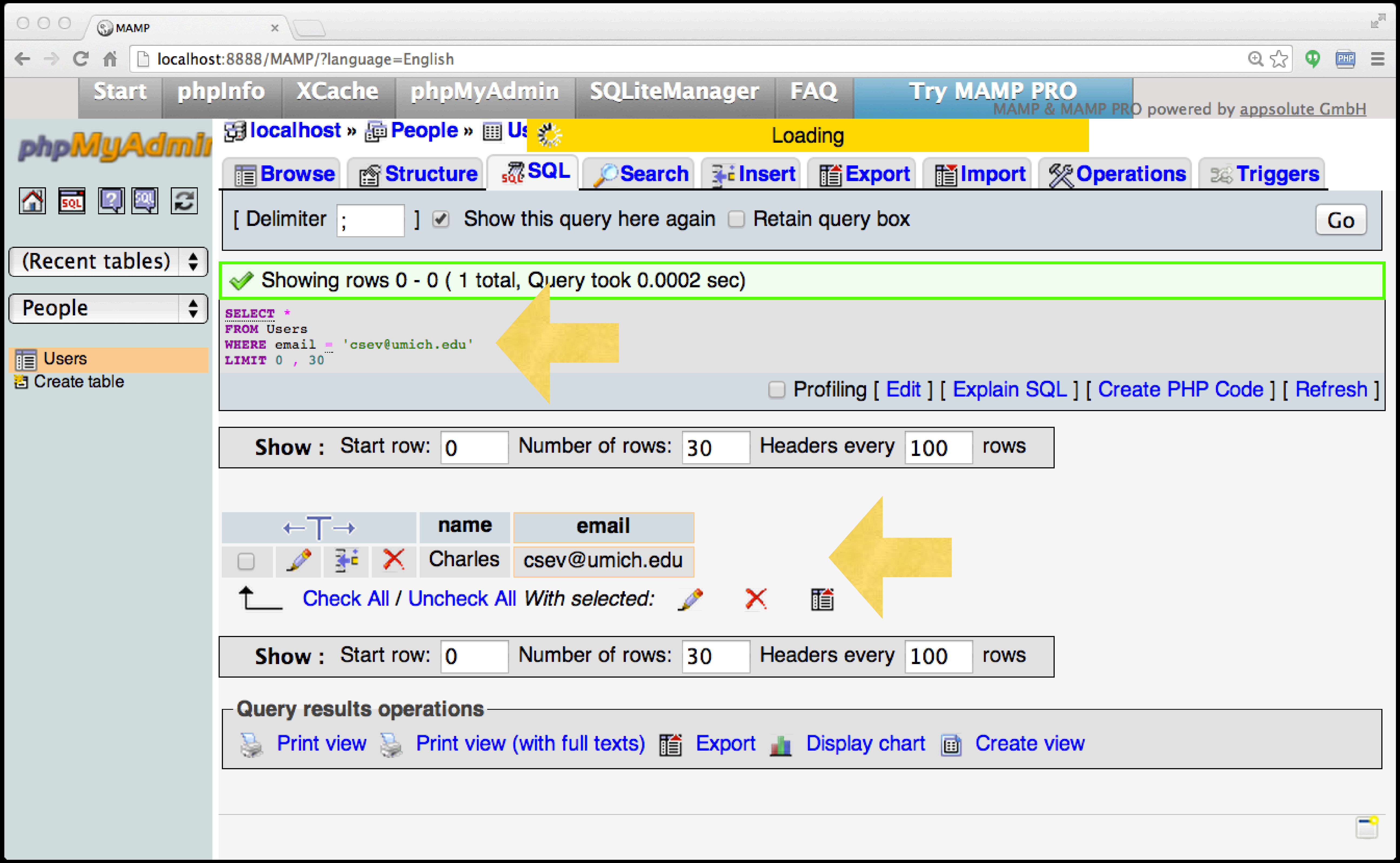
Retrieving Records: Select

Retrieves a group of records - you can either retrieve all the records or a subset of the records with a WHERE clause

```
SELECT * FROM Users
```

```
SELECT * FROM Users WHERE email='csev@umich.edu'
```

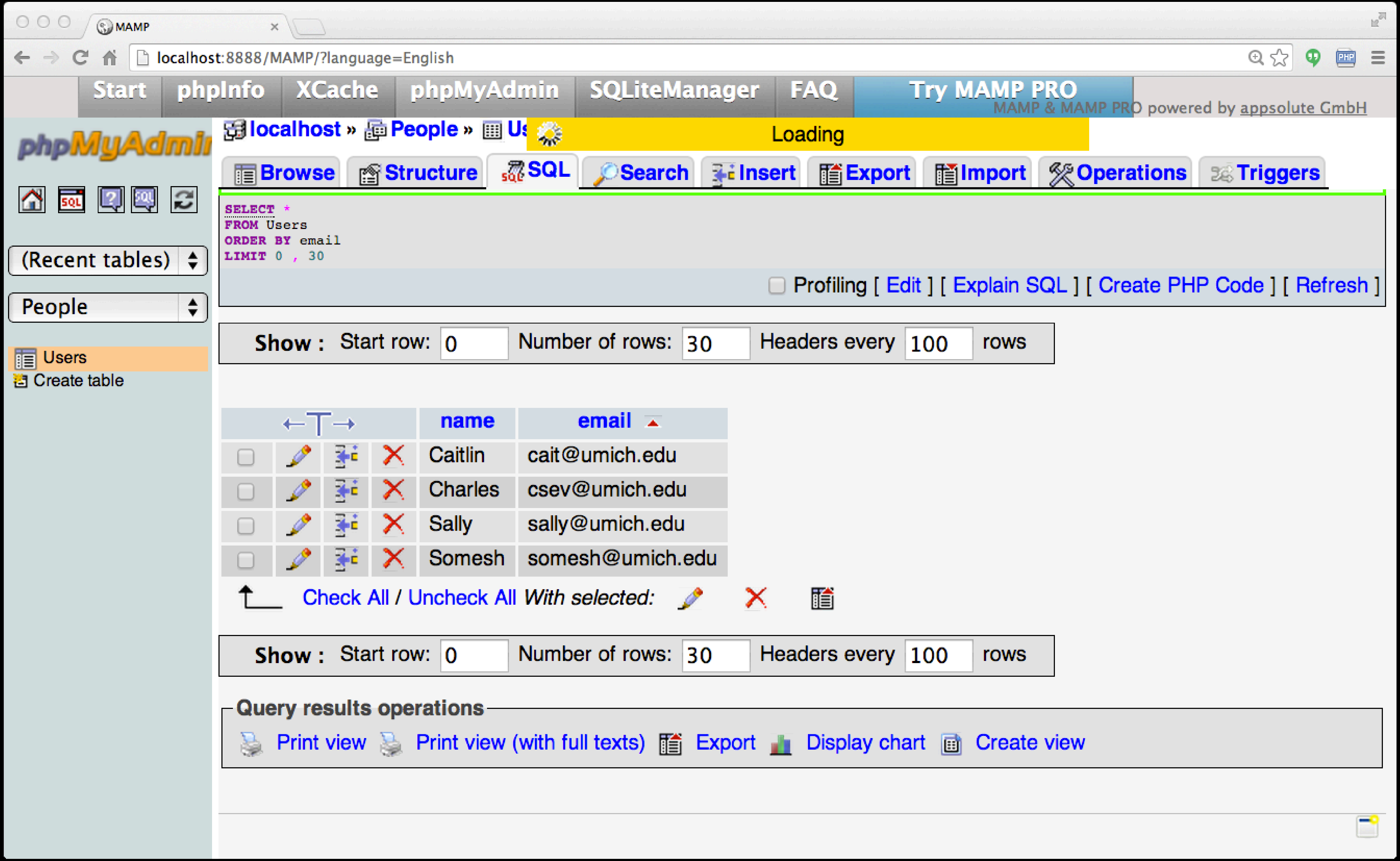

MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
localhost » People » Us
Loading
Browse
Structure
SQL
Search
Insert
Export
Import
Operations
Triggers
(Recent tables)
People
Users
Create table
Showing rows 0 - 3 (4 total, Query took 0.0002 sec)
SELECT *
FROM Users
LIMIT 0 , 30
Profiling [Edit] [Explain SQL] [Create PHP Code] [Refresh]
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
name
email
Charles
csev@umich.edu
Sally
sally@umich.edu
Somesh
somesh@umich.edu
Caitlin
cait@umich.edu
Check All / Uncheck All With selected:
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
Query results operations
Print view
Print view (with full texts)
Export
Display chart
Create view


MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
localhost » People » Us
Loading
Browse
Structure
SQL
Search
Insert
Export
Import
Operations
Triggers
[Delimiter ;] Show this query here again Retain query box
Go
(Recent tables)
People
Users
Create table
Showing rows 0 - 0 (1 total, Query took 0.0002 sec)
SELECT *
FROM Users
WHERE email = 'csev@umich.edu'
LIMIT 0 , 30
Profiling [Edit] [Explain SQL] [Create PHP Code] [Refresh]
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
name
email
Charles
csev@umich.edu
Check All / Uncheck All With selected:
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
Query results operations
Print view
Print view (with full texts)
Export
Display chart
Create view

Sorting with ORDER BY

You can add an ORDER BY clause to SELECT statements to get the results sorted in ascending or descending order

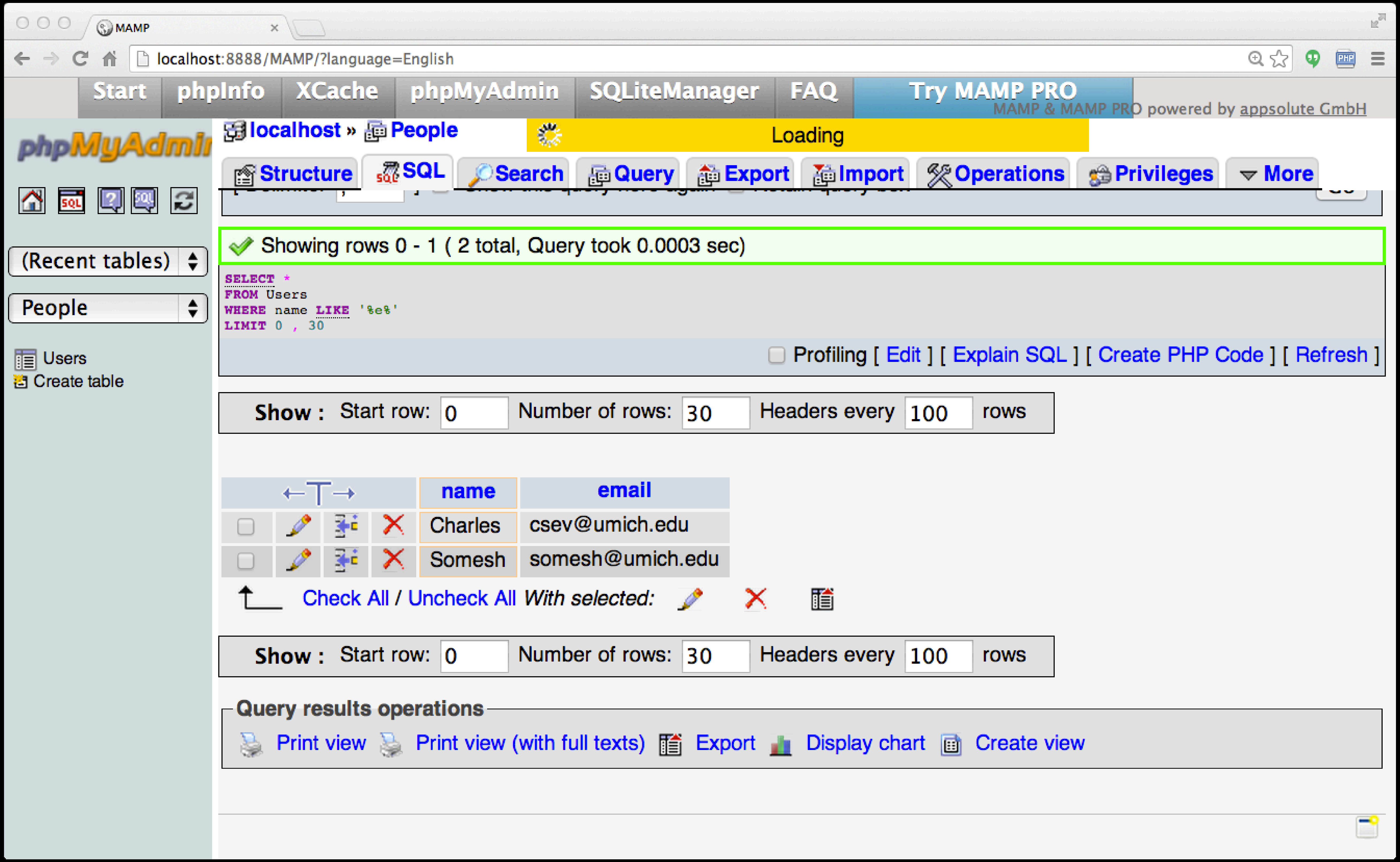
```
SELECT * FROM Users ORDER BY email
```


MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
localhost » People » Us
Loading
(Recent tables)
People
Users
Create table
Browse
Structure
SQL
Search
Insert
Export
Import
Operations
Triggers
SELECT *
FROM Users
ORDER BY email
LIMIT 0 , 30
Profiling [Edit] [Explain SQL] [Create PHP Code] [Refresh]
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
name
email
Caitlin cait@umich.edu
Charles csev@umich.edu
Sally sally@umich.edu
Somesh somesh@umich.edu
Check All / Uncheck All With selected:
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
Query results operations
Print view
Print view (with full texts)
Export
Display chart
Create view

The LIKE Clause

We can do wildcard matching in a WHERE clause using the LIKE operator

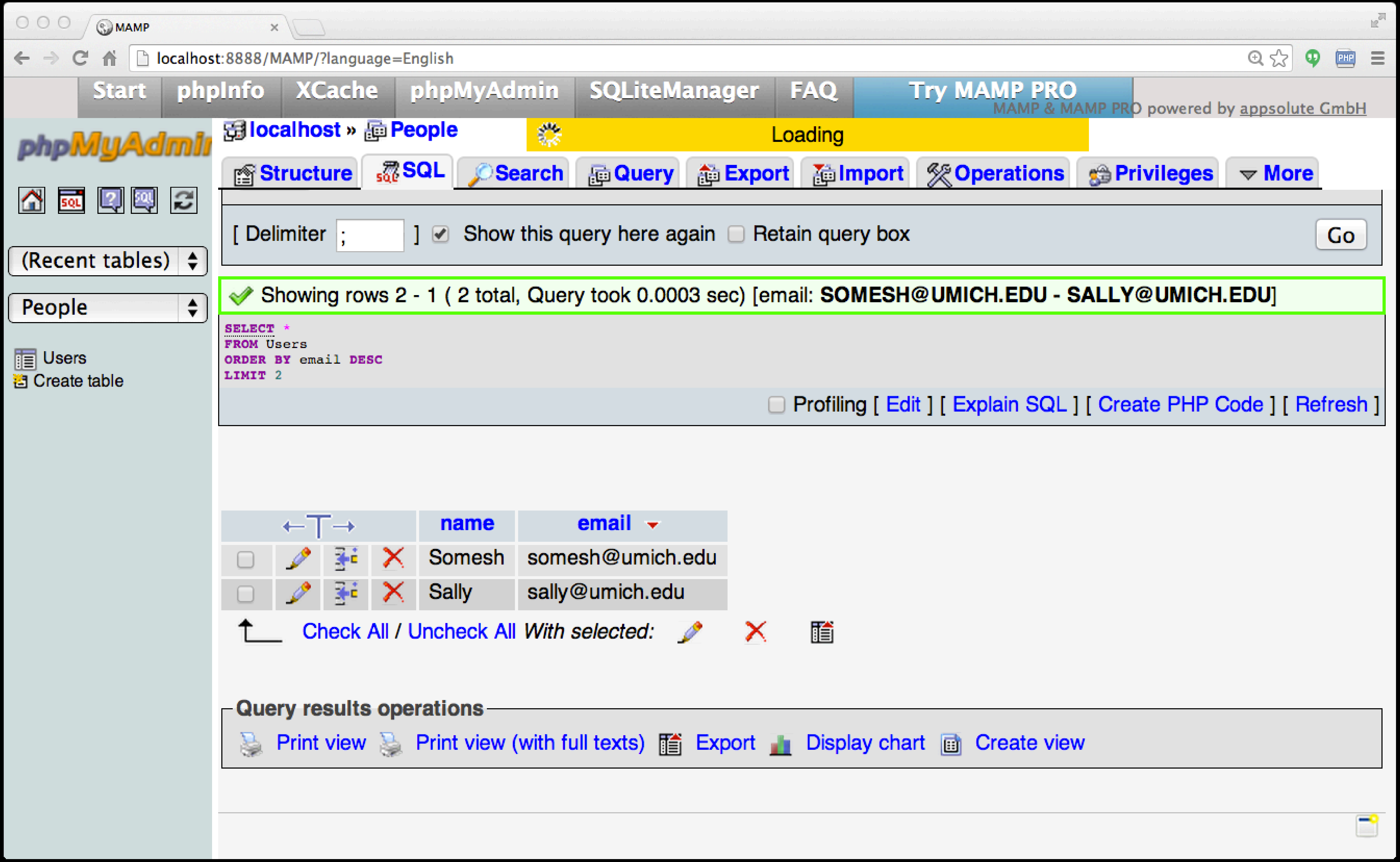
```
SELECT * FROM Users WHERE name LIKE '%e%'
```

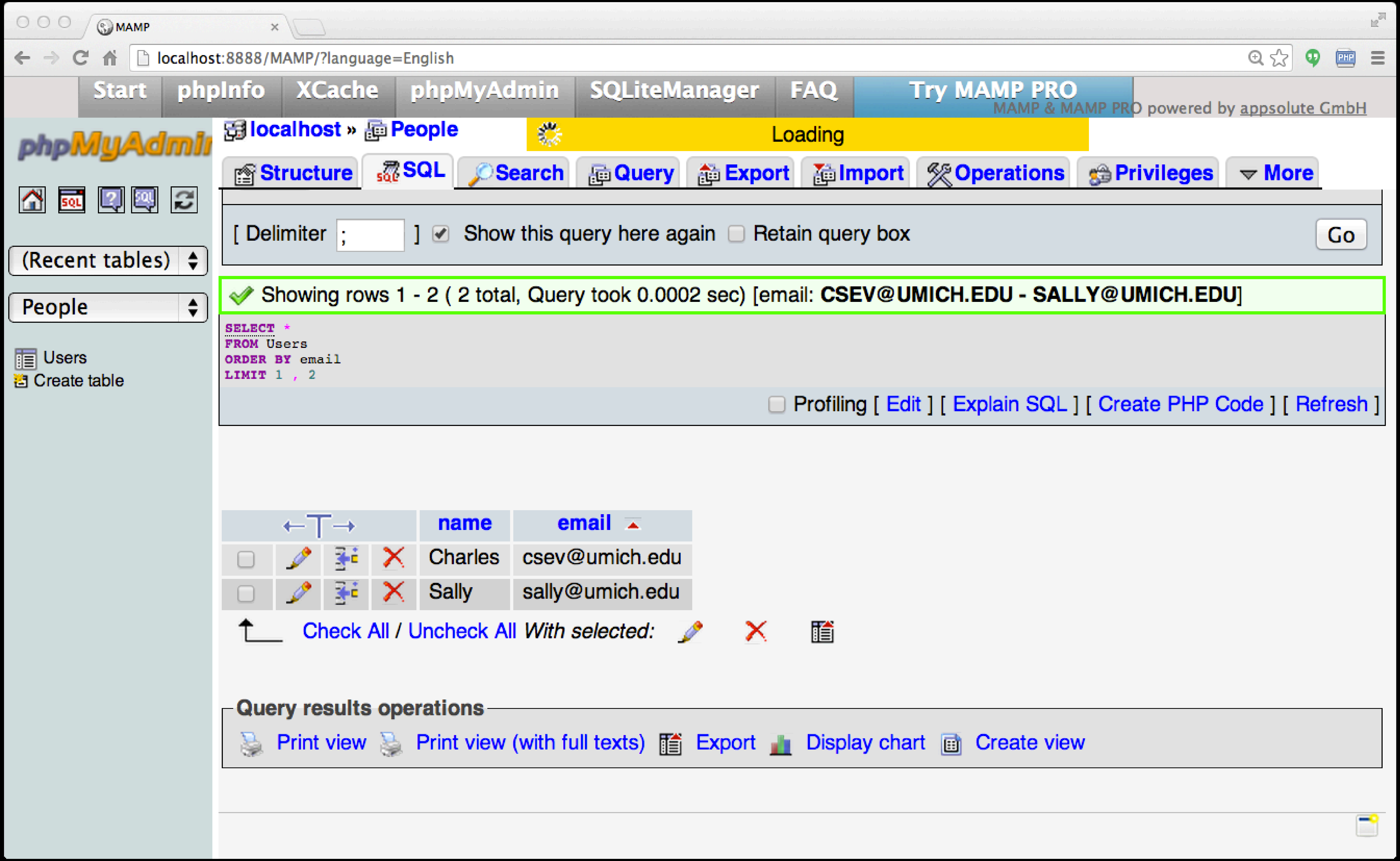

MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
localhost » People
Loading
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
More
(Recent tables)
People
Users
Create table
Showing rows 0 - 1 (2 total, Query took 0.0003 sec)
SELECT *
FROM Users
WHERE name LIKE '%e%'
LIMIT 0 , 30
Profiling [Edit] [Explain SQL] [Create PHP Code] [Refresh]
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
name
email
Charles
csev@umich.edu
Somesh
somesh@umich.edu
Check All / Uncheck All With selected:
Show : Start row: 0 Number of rows: 30 Headers every 100 rows
Query results operations
Print view
Print view (with full texts)
Export
Display chart
Create view

The LIMIT Clause

- The LIMIT clause can request the first "n" rows, or the first "n" rows after some starting row. Note: the first row is zero, not one.
- WHERE and ORDER BY clauses happen *before* the LIMIT is applied.
- The limit can be a count or a starting row and count (starts from 0).

```
SELECT * FROM Users ORDER BY email DESC LIMIT 2;
SELECT * FROM Users ORDER BY email LIMIT 1,2;
```


MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
localhost » People
Loading
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
More
(Recent tables)
People
Users
Create table
[Delimiter ;] Show this query here again Retain query box
Go
Showing rows 2 - 1 (2 total, Query took 0.0003 sec) [email: SOMESH@UMICH.EDU - SALLY@UMICH.EDU]
SELECT *
FROM Users
ORDER BY email DESC
LIMIT 2
Profiling [Edit] [Explain SQL] [Create PHP Code] [Refresh]
name
email
Somesh
somesh@umich.edu
Sally
sally@umich.edu
Check All / Uncheck All With selected:
Query results operations
Print view
Print view (with full texts)
Export
Display chart
Create view


MAMP
localhost:8888/MAMP/?language=English
Start
phpInfo
XCache
phpMyAdmin
SQLiteManager
FAQ
Try MAMP PRO
MAMP & MAMP PRO powered by appsolute GmbH
phpMyAdmin
localhost » People
Loading
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
More
(Recent tables)
People
Users
Create table
[Delimiter ;] Show this query here again Retain query box
Go
Showing rows 1 - 2 (2 total, Query took 0.0002 sec) [email: CSEV@UMICH.EDU - SALLY@UMICH.EDU]
SELECT *
FROM Users
ORDER BY email
LIMIT 1 , 2
Profiling [Edit] [Explain SQL] [Create PHP Code] [Refresh]
name
email
Charles
csev@umich.edu
Sally
sally@umich.edu
Check All / Uncheck All With selected:
Query results operations
Print view
Print view (with full texts)
Export
Display chart
Create view

Counting Rows with SELECT

You can request to receive the count of the rows that would be retrieved instead of the rows

```
SELECT COUNT(*) FROM Users;
SELECT COUNT(*) FROM Users WHERE email='csev@umich.edu'
```

SQL Summary

```
INSERT INTO Users (name, email) VALUES ('Ted', 'ted@umich.edu')

DELETE FROM Users WHERE email='ted@umich.edu'

UPDATE Users SET name='Charles' WHERE email='csev@umich.edu'

SELECT * FROM Users WHERE email='csev@umich.edu'

SELECT * FROM Users ORDER BY email

SELECT * FROM Users WHERE name LIKE '%e%'

SELECT * FROM Users ORDER BY email LIMIT 1,2;

SELECT COUNT(*) FROM Users WHERE email='csev@umich.edu'
```

This is not too exciting (so far)

- Tables pretty much look like big, fast programmable spreadsheets with rows, columns, and commands.
- The power comes when we have more than one table and we can exploit the relationships between the tables.

Data Types in SQL

Looking at Data Types

- Text fields (small and large)
- Binary fields (small and large)
- Numeric fields
- AUTO_INCREMENT fields

String Fields

- Understand character sets and are indexable for searching
- CHAR allocates the entire space (faster for small strings where length is known)
- VARCHAR allocates a variable amount of space depending on the data length (less space)

Text Fields

- Have a character set - paragraphs or HTML pages
  - TINYTEXT up to 255 characters
  - TEXT up to 65K
  - MEDIUMTEXT up to 16M
  - LONGTEXT up to 4G
- Generally not used with indexing or sorting - and only then limited to a prefix

Binary Types (rarely used)

- Character = 8 - 32 bits of information depending on character set
- Byte = 8 bits of information
  - BYTE(n) up to 255 bytes
  - VARBINARY(n) up to 65K bytes
- Small Images - data
- Not indexed or sorted

Binary Large Object (BLOB)

- Large raw data, files, images, word documents, PDFs, movies, etc.
- No translation, indexing, or character set
  - TINYBLOB(n) - up to 255
  - BLOB(n) - up to 65K
  - MEDIUMBLOB(n) - up to 16M
  - LONGBLOB(n) - up to 4G

Integer Numbers

Integer numbers are very efficient, take little storage, and are easy to process because CPUs can often compare them with a single instruction.

- TINYINT (-128, 128)
- SMALLINT (-32768, +32768)
- INT or INTEGER (2 Billion)
- BIGINT - (10**18 ish)

Floating Point Numbers

Floating point numbers can represent a wide range of values, but accuracy is limited.

- FLOAT (32-bit) 10**38 with seven digits of accuracy
- DOUBLE (64-bit) 10**308 with 14 digits of accuracy

Dates

- TIMESTAMP - 'YYYY-MM-DD HH:MM:SS' (1970, 2037)
- DATETIME - 'YYYY-MM-DD HH:MM:SS'
- DATE - 'YYYY-MM-DD'
- TIME - 'HH:MM:SS'
- Built-in MySQL function NOW()

Database Keys and Indexes

AUTO_INCREMENT

Often as we make multiple tables and need to JOIN them together we need an integer primary key for each row so we can efficiently add a reference to a row in some other table as a foreign key.

```
DROP TABLE Users;

CREATE TABLE Users (
  user_id INT UNSIGNED NOT NULL
    AUTO_INCREMENT,
  name VARCHAR(128),
  email VARCHAR(128),
  PRIMARY KEY(user_id),
  INDEX(email)
)
```

MySQL Functions

Many operations in MySQL need to use the built-in functions (like NOW() for dates).

- http://dev.mysql.com/doc/refman/5.0/en/string-functions.html
- http://dev.mysql.com/doc/refman/5.0/en/date-and-time-functions.html

Indexes

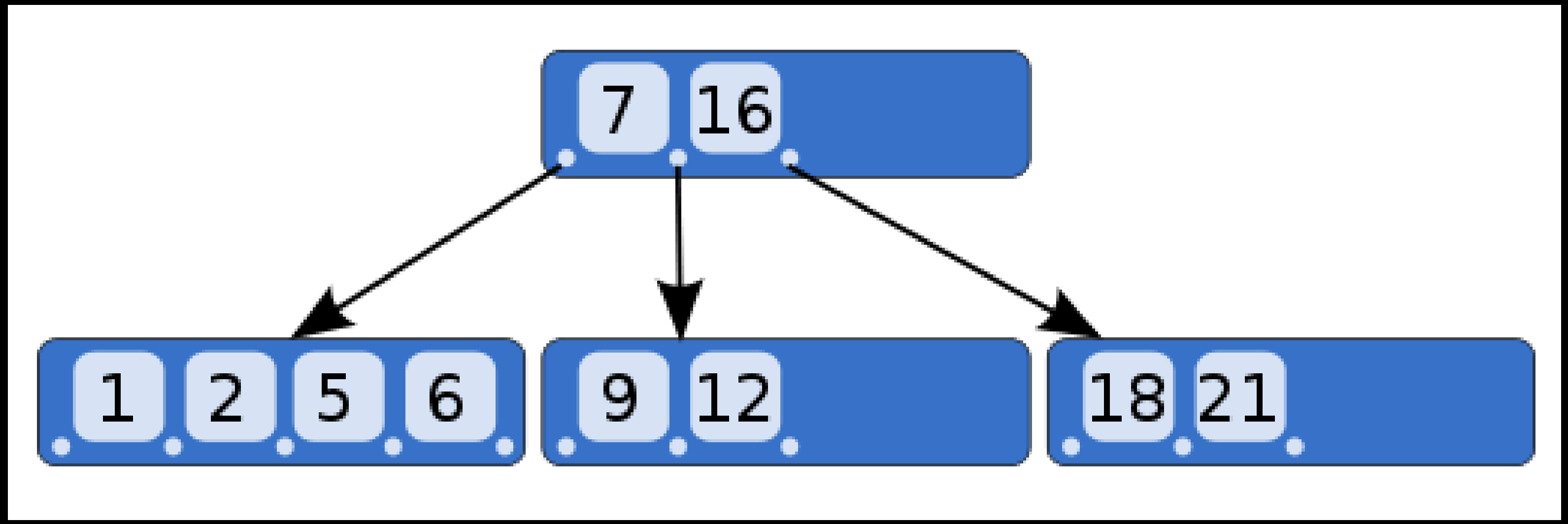
- As a table gets large (they always do), scanning all the data to find a single row becomes very costly
- When drchuck@gmail.com logs into FaceBook, they must find my password amongst 500 million users
- There are techniques to greatly shorten the scan as long as you create data structures and maintain those structures - like shortcuts
- Hashes or Trees

MySQL Index Types

- PRIMARY KEY - Very little space, exact match, requires no duplicates, extremely fast for integer fields
- INDEX - Good for individual row lookup and sorting / grouping results - works best with exact matches or prefix lookups - can suggest HASH or BTREE

B-Trees

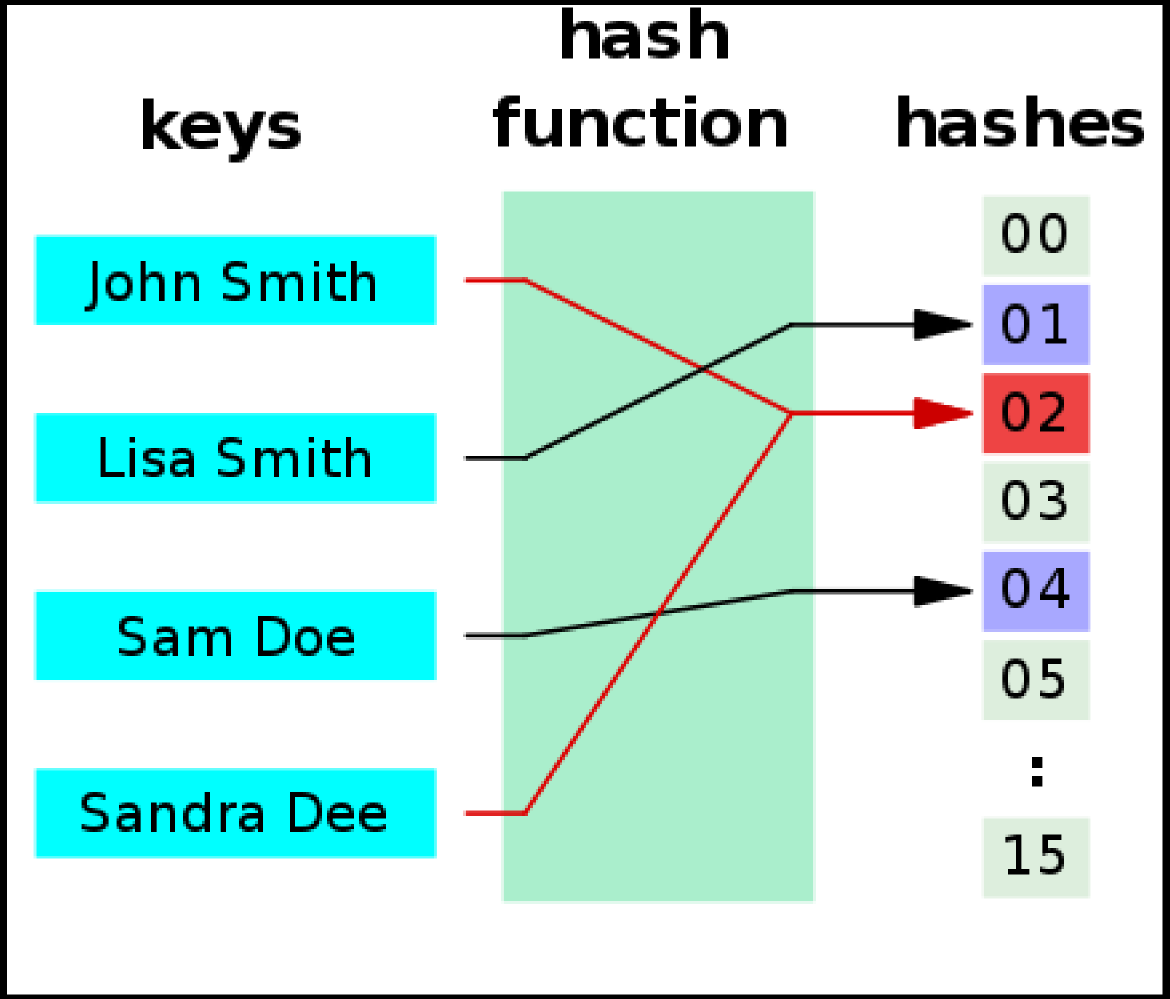
A B-tree is a tree data structure that keeps data sorted and allows searches, sequential access, insertions, and deletions in logarithmic amortized time. The B-tree is optimized for systems that read and write large blocks of data. It is commonly used in databases and file systems.



http://en.wikipedia.org/wiki/B-tree

Hashes

A hash function is any algorithm or subroutine that maps large data sets to smaller data sets, called keys. For example, a single integer can serve as an index to an array (cf. associative array). The values returned by a hash function are called hash values, hash codes, hash sums, checksums, or simply hashes. Hash functions are mostly used to accelerate table lookup or data comparison tasks such as finding items in a database...



http://en.wikipedia.org/wiki/Hash_function

Specifying Indexes

```
DROP TABLE Users;

CREATE TABLE Users (
  user_id INT UNSIGNED NOT NULL
    AUTO_INCREMENT,
  name VARCHAR(128),
  email VARCHAR(128),
  PRIMARY KEY(user_id),
  INDEX(email)
)
```

```
ALTER TABLE Users ADD INDEX ( email ) USING BTREE
```

Summary

- SQL allows us to describe the shape of data to be stored and give many hints to the database engine as to how we will be accessing or using the data.
- SQL is a language that provides us operations to Create, Read, Update, and Delete (CRUD) our data in a database.

Acknowledgements / Contributions

Initial Development: Charles Severance, University of Michigan School of Information

Insert new Contributors and Translators here including names and dates

Continue new Contributors and Translators here